

TUSHAR JAIN

Co-Founder · Full Stack Developer · Embedded Systems Engineer · MVP Builder

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SUMMARY

Engineering student at BMS College of Engineering (CSBS, 2024 - 2028) with a track record of shipping real-world products from a health-tech wearable pitched at JIC Accelerator 7.0, to a metro app with 1000 Play Store downloads, to a B2B SaaS reservation system. Works across the full stack: React/Next.js frontends, Supabase and firebase backends, ESP32/Arduino embedded firmware, and BLE hardware. Driven by a philosophy of utility over ornamentation every build solves a real user problem.

EXPERIENCE

Cofounder & Tech Lead -- PulsePredict (Vital Health Tech)

Aug 2025 – Jan 2026

Jaipur, India · Pitched at JIC Accelerator 7.0 · tusharjain.in/work/vital-health-tech

- Identified a critical gap in India's geriatric care market fall detection & vitals monitoring devices priced above ₹10,000 excluded the mass market. Conceptualized a sub-₹6,000 solution targeting 140M+ elderly Indians.
- Solved critical false-positive challenges in fall detection through iterative motion threshold tuning during user testing, achieving a reliable alert system.
- Validated business model against 3 direct competitors at JIC Accelerator 7.0; secured industry feedback that shaped the product roadmap and sub-62,000 pricing strategy.
- Designed architecture for future ABDM (National Digital Health Mission) integration, enabling scale into India's official digital health infrastructure.
- Tech Stack: ESP32, MPU6050, MAX30102 (SpO2), BLE/Bluetooth, C++, Embedded Firmware, Android (APK Build & Publish).

Web Development Intern - 3Skill Training

- Completed a structured web development internship covering frontend fundamentals, responsive design, and deployment workflows. [View Certificate](#)

PROJECTS

1. Jaipur Ride (Metro) Public Utility Smart Transport Android

[Live Demo](#) · [Google Play Store](#) · [GitHub](#) · [Project Page](#)

- Developed a public utility Android application for Jaipur Metro, providing route planning, fare calculation, station information, and train timings through a single, easy-to-use platform.
- Achieved **1,000+ Google Play Store downloads** with **250+ monthly active users (MAU)**, helping commuters access reliable metro information without relying on scattered online sources.
- Designed a lightweight transit data architecture to efficiently manage route nodes, fare calculations, and schedule data, ensuring fast performance even on low-connectivity networks.
- Built a responsive, mobile-first interface focused on speed, accessibility, and an intuitive commuter experience, with continuous improvements based on real user feedback.
- **Tech Stack:** React.js, JavaScript (ES6+), HTML5, CSS3, Vercel.

2. NotesCSBS Productivity Academic Notes Platform

[Live Demo](#) · [Project Page](#)

- Developed a centralized academic platform for **BMSCE CSBS students**, providing semester-wise notes, lab materials, previous year papers, and study resources through a single searchable portal.
- Built a production-ready full-stack application with **React.js**, **Supabase Authentication**, **SQL-backed database**, secure file management, an **admin dashboard**, and Google Drive integration for scalable content management.
- Achieved **500+ organic Google Search clicks**, **3K+ search impressions**, **16% click-through rate (CTR)**, and an **average Google search position of 4.7** through SEO optimization and structured content.
- Designed a responsive, mobile-first interface optimized for high-traffic examination periods, enabling students to quickly discover and access curated academic resources.
- **Tech Stack:** React.js, JavaScript, Supabase (Auth & PostgreSQL), SQL, Google Drive Integration, HTML5, CSS3, Vercel, Google Search Console (SEO).

3. Namma Ride (Bengaluru Metro) Public Utility UX Multilingual

[Project Page](#) · tusharjain.in/projects

- Bengaluru Metro journey planner with full support for English, Hindi, and Kannada ensuring accessibility for the city's diverse demographic.
- Built with vanilla JavaScript to maintain an ultra-lightweight footprint for users on weak cellular networks; complex journey-planner logic with multilingual state transitions.
- Coming soon to Android Google Play Store; building on learnings from Jaipur Ride's successful Play Store launch.
- Tech Stack: HTML, CSS, JavaScript (Vanilla).

4. Split-Payment System FinTech System Design Prototype

[GitHub](#) · [LinkedIn Post](#) · [Project Page](#)

- Concept-level fintech system enabling a single purchase to be split across multiple payment sources simultaneously (e.g., UPI + Credit Card).
- Designed transaction atomicity logic: if any one payment source fails, the entire transaction is safely rolled back no partial charges or duplicate deductions.
- Integrated Razorpay test mode to simulate real payment gateway interactions; explored edge cases including retries, timeout failures, and partial authorization scenarios.
- Tech Stack: JavaScript, Node.js, Razorpay (Test Mode Integration).

5. BookMySlot B2B SaaS Operations Real time

[Live Demo](#) · [GitHub](#) · [Project Page](#)

- Verticalized B2B SaaS reservation system targeting Indian small businesses (restaurants as BookMyDine, clinics as BookMyClinic) that still operate on paper, WhatsApp, and phone calls.
- Designed a full system not just one app with three synchronized touchpoints: customer-facing reservation website, staff/waiter live sync app, and owner admin panel with full operational visibility.
- Used Supabase real-time database to sync state instantly across all three interfaces; Google Sheets API for lightweight owner reporting.
- Focused on Indian business reality: ultra-low cost, mobile-first, and simple enough for non-technical staff to operate without training.
- Tech Stack: HTML/CSS/JS, Supabase (Auth + Real-time DB), firebase, Vercel, Google Sheets API.

6. Air Guitar Embedded Systems Audio Synthesis Hardware

[GitHub](#) · [LinkedIn Post](#) · [Project Page](#)

- Gesture-controlled virtual guitar that synthesizes string sounds in real-time using the Karplus-Strong algorithm no pre-recorded samples, making it feel like a real physical instrument.
- MPU6050 IMU tracks wrist tilt (string selection) and flicking acceleration (strumming velocity); data streamed over Serial at 115200 baud to a Python audio engine.
- Achieved sub-10ms latency by implementing thread-safe audio callbacks; applied delta-based calibration to mitigate Serial data jitter for a responsive 'strum' feel.
- Velocity-sensitive logic adjusts synthesized sound based on acceleration magnitude, creating dynamic playing expression.
- Tech Stack: Python, Arduino, MPU6050 (IMU), NumPy, Karplus-Strong Synthesis, Serial Communication (115200 baud).

7. Indigo In-Flight - Air-Stream Systems Offline First Web

[GitHub](#) · [Project Page](#)

- Offline-first in-flight entertainment web system inspired by how airlines stream content without internet built to explore air-gapped local network constraints.
- Designed fully offline-capable frontend with no external CDN dependencies; assets load reliably over a local server without internet access.
- Architected for multi-device delivery (phones, tablets) over a local network simulating real airline seatback entertainment delivery infrastructure.
- Key insight: offline systems require fundamentally different UX assumption's reliability and predictability matter more than feature richness.
- Tech Stack: HTML, CSS, JavaScript, Offline-first architecture.

CERTIFICATIONS

- Investment Risk Management - [Coursera](#) (2025)
- Business Analysis & Process Management - [Coursera](#) (2025)

ACHIEVEMENTS

- TOP 10 FINALIST , CONFERENCE 2.0 HACKATHON (2026)
selected among 500+ participating teams for developing safe route, an ai-powered safer route planning solution.
- JIC ACCELERATOR 7.0 - pitched pulse predict, a low-cost embedded health-tech wearable for elderly safety and remote monitoring.
- JAIPUR RIDE - reached 1,000+ google play store downloads and 250+ monthly active users (MAU).

RESEARCH

Human-Supervised Dual-UUV System for Autonomous Maritime Surveillance [Research Paper](#) · tusharjain.in/research

Autonomous Systems · Sensor Fusion · Human in the Loop Control · EKF Localization

- Proposed a split-architecture dual-UUV surveillance framework combining autonomous acoustic threat detection (passive UUV) with operator-controlled inspection (active UUV).
- Utilized Extended Kalman Filters (EKF) for precise underwater localization, accounting for detection range limits and underwater acoustic communication constraints.
- Defined a strict Human in the Loop protocol for all active system responses ensuring safety, legal compliance, and ethical operation in maritime security contexts.
- Analyzed theoretical models for localization accuracy, communication range constraints, and coordination latency between dual-vehicle systems.

EDUCATION

Bachelor of Engineering - Computer Science & Business Systems (CSBS)

2024 – 2028

BMS College of Engineering (BMSCE) · Bengaluru, India **CGPA: 9.08 / 10.0** (up to 4rd Semester)

SKILLS & TECHNOLOGIES

Frontend: React.js, Next.js, HTML5, CSS3, JavaScript (ES6+), Vanilla JS, Responsive Design, Mobile-first UI

Backend & Database: Supabase (Auth + Real-time DB), Firebase, Google Sheets API, Node.js, REST APIs

Hardware & Embedded: ESP32, Arduino, MPU6050 (IMU), MAX30102 (SpO2), BLE/Bluetooth, I2C/UART/SPI, Sensor Fusion, C/C++ Firmware, MATLAB (academic).

Programming Languages: JavaScript, Python (Academic), MATLAB (Academic), C, C++

Tools & Platforms: Git/GitHub, Vercel, Razor pay (Test Mode), Android (APK Build & Publish), Antigravity, VS Code

Areas: System Design, Distributed Transactions, Offline-first Architecture, Real-time Sync, Edge Case Management, MVP Strategy

Portfolio: tusharjain.in · All Projects: tusharjain.in/projects · GitHub: github.com/Tusharjain-19 · LinkedIn: in/tushar-jain-781149322